

When and How Might You Die?

Visualizing Age & Cause of Death using UN & WHO Data

Link to Viz: <https://observablehq.com/d/f63cf70b0a1eee0e>

Problem & Motivation

“When and how will I die?”

Problem with life expectancy & general cause-of-death stats:

- Feels too distant
- Does not show the full picture of risks over lifetime

Life Expectancy

[Print](#)

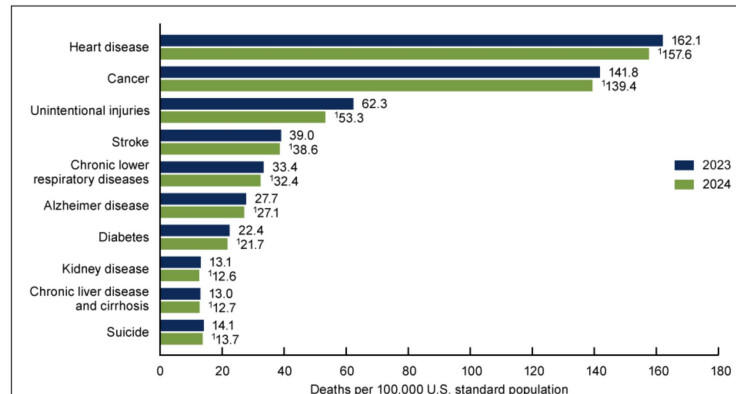
Data are for the U.S.

Life expectancy at birth

- Both sexes: 79.0 years
- Males: 76.5 years
- Females: 81.4 years

[Source](#)

Figure 4. Age-adjusted death rate for the 10 leading causes of death in 2024: United States, 2023 and 2024



[Source](#)

Problem & Motivation

“When and how will I die?”

Goals:

- Turn population data into individual experience
- Show risks individuals experience across different stages of life

Problem & Motivation

“When and how will I die?”

Proposed Approach:

- Use population data to simulate when and how people die
 - Inspired by <https://flowingdata.com/2015/09/23/years-you-have-left-to-live-probably/>
- Visualize both the process and results

Datasets

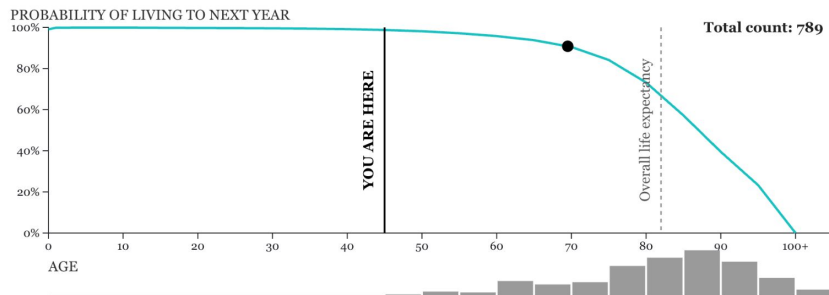
- United Nations (UN) mortality statistics
 - <https://population.un.org/dataportal/home?df=9456c5d3-9e5a-409a-a65d-1703294f95bb>
- World Health Organization (WHO) causes of death data
 - <https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates/ghe-leading-causes-of-death>

Key Visualization

Overall View

When and How Might I Die?

I am and currently years old living in .

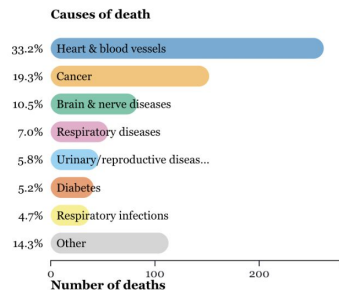
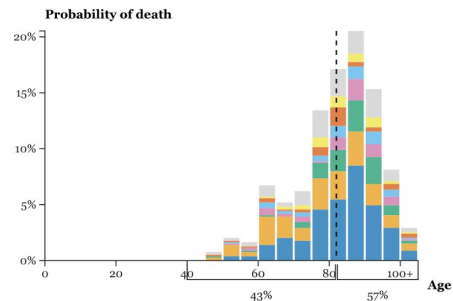


Animation speed

☐ slow ☒ fast

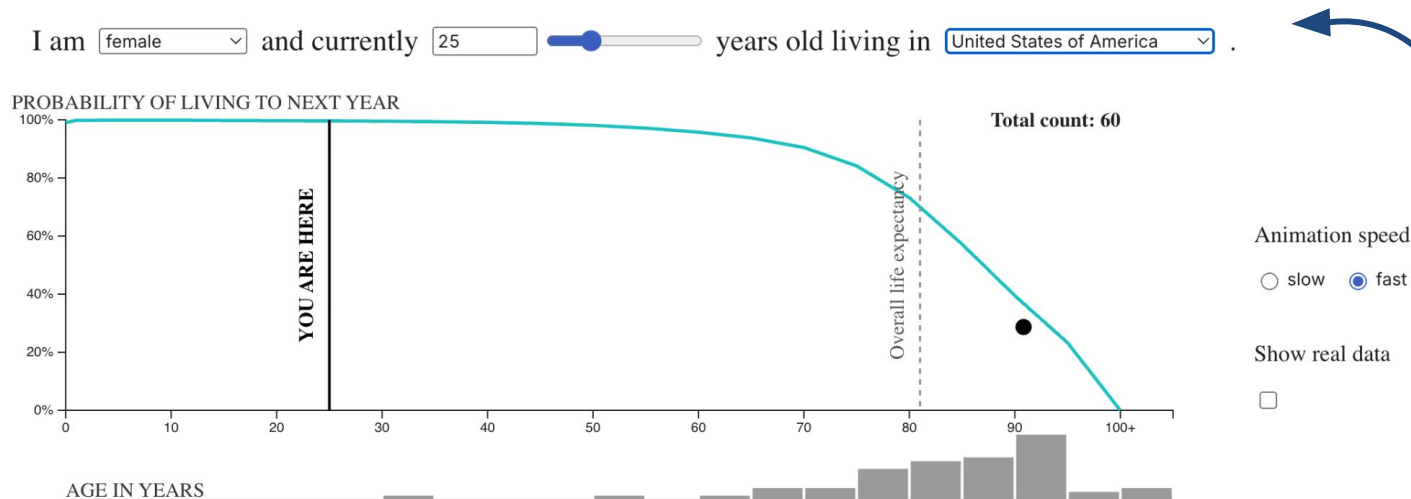
Show real data

☐



Key Visualization

User selection

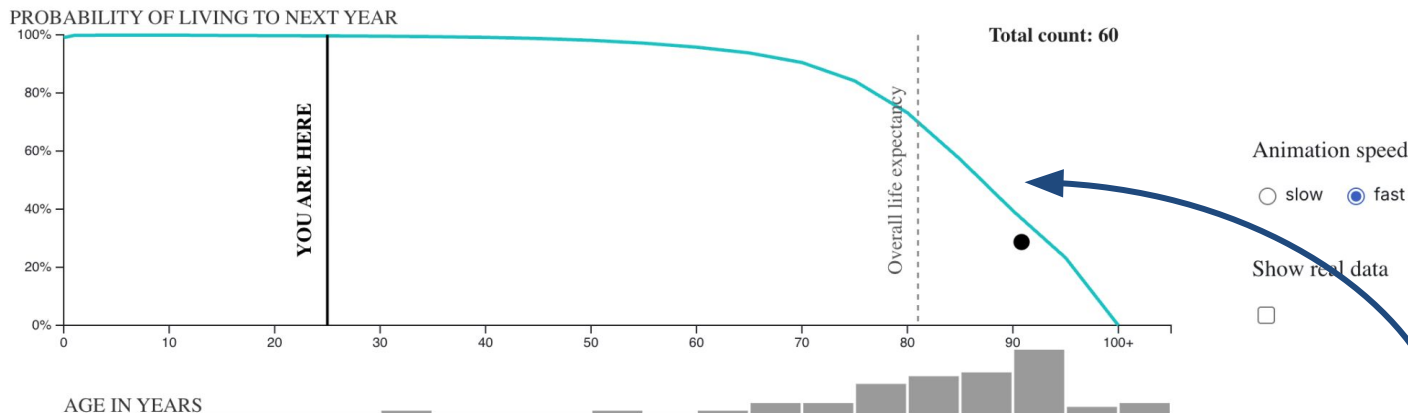


Users select their gender, current age, and country

Key Visualization

Survival Probability Chart

I am and currently years old living in .



Green curve shows corresponding survival probability across ages

Key Visualization

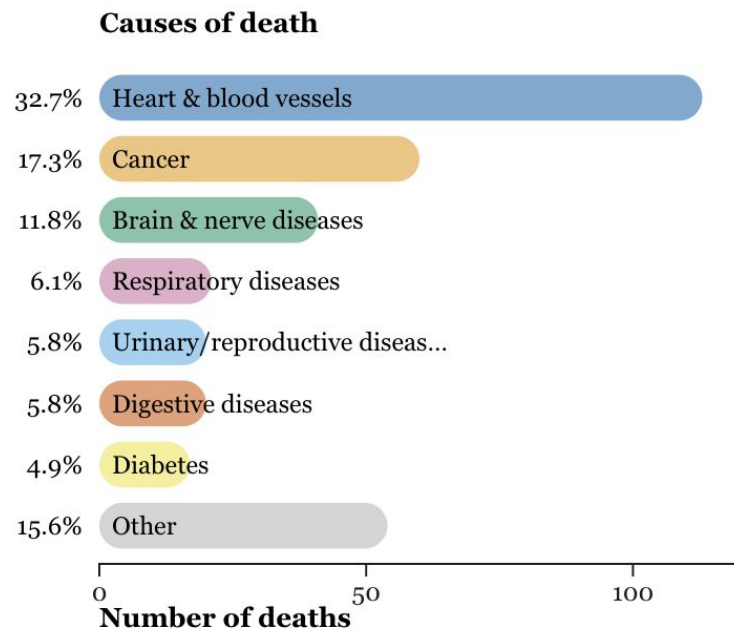
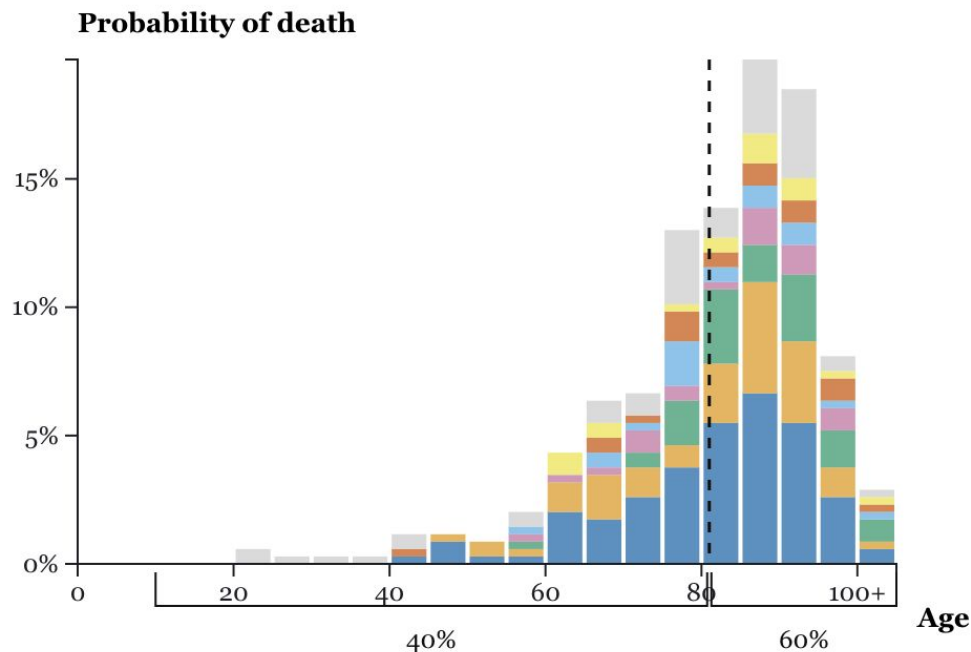
Animated simulation

See demo video

- A moving dot represents one simulated life
- Starts at the current age
- Moves along the survival probability curve
- At each age, death is determined by the probability of survival
- When death occurs, the dot falls
- After death:
 - A cause of death is assigned
 - Based on gender, region, and age
- A new dot appears and the simulation repeats

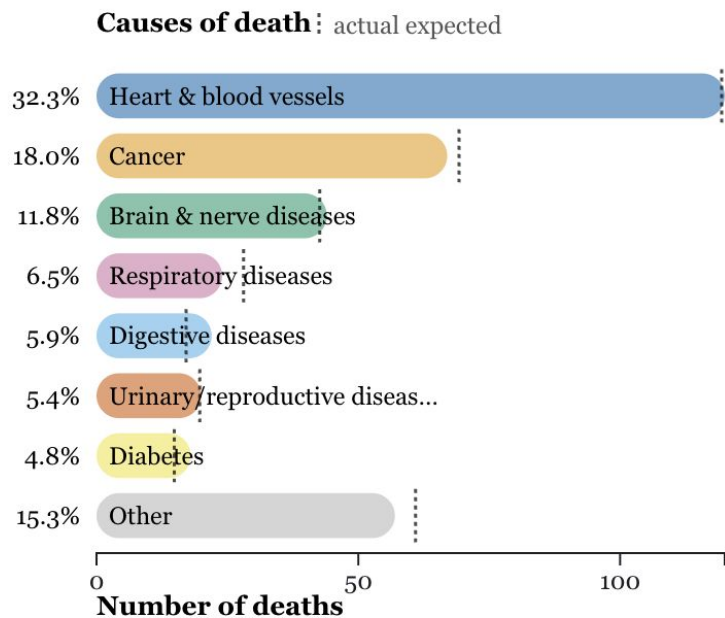
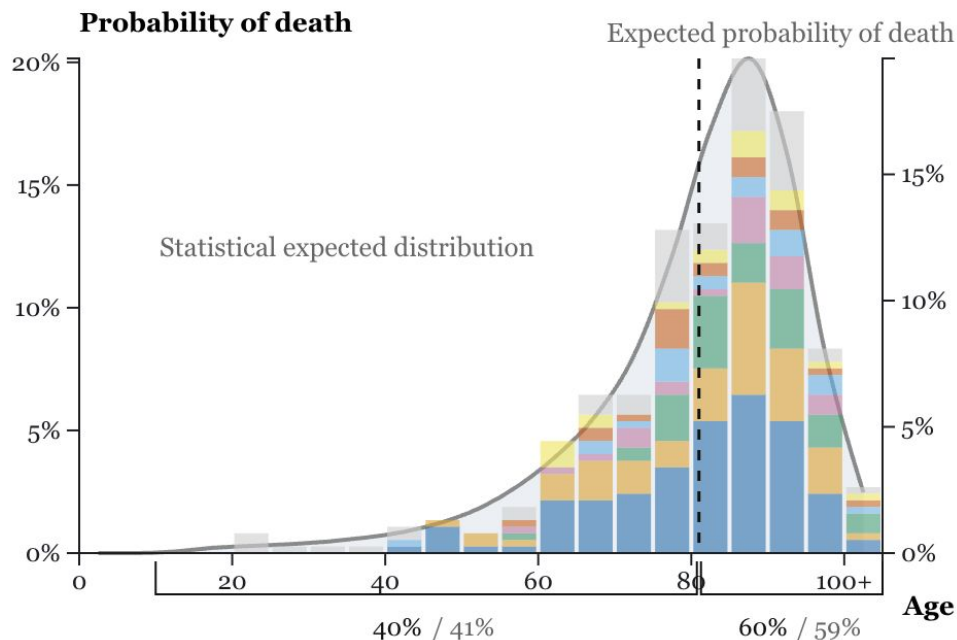
Key Visualization

Death age distribution & Cause-of-death counts



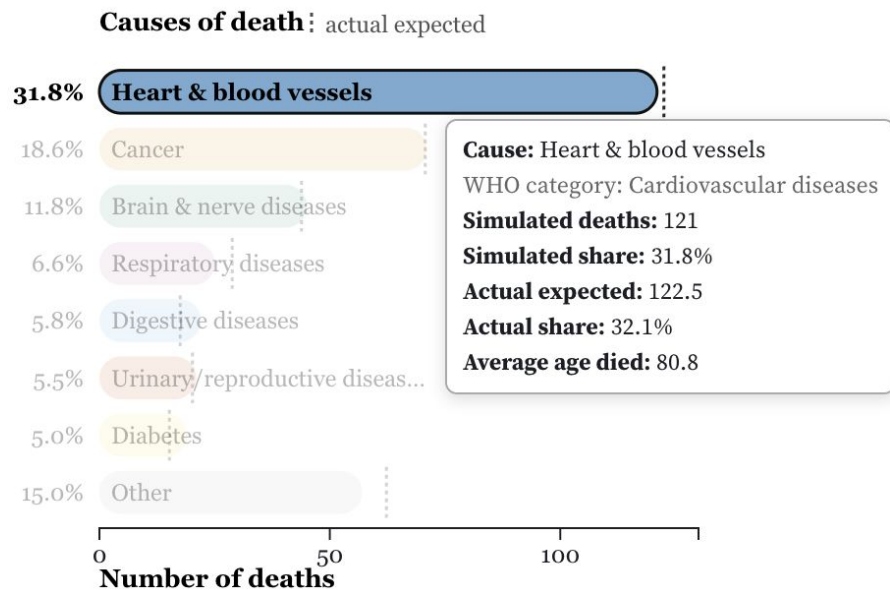
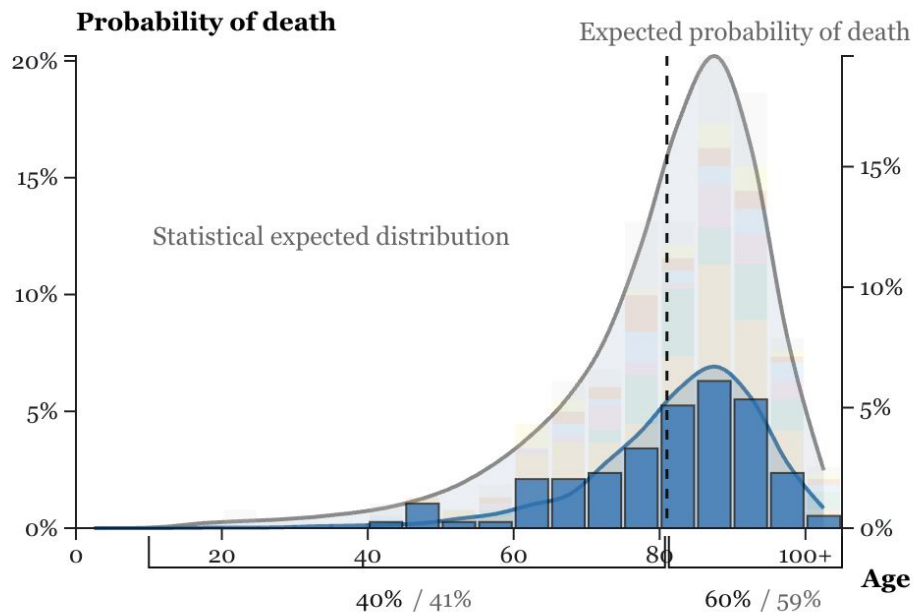
Key Visualization

Display real data on top of simulated results



Key Visualization

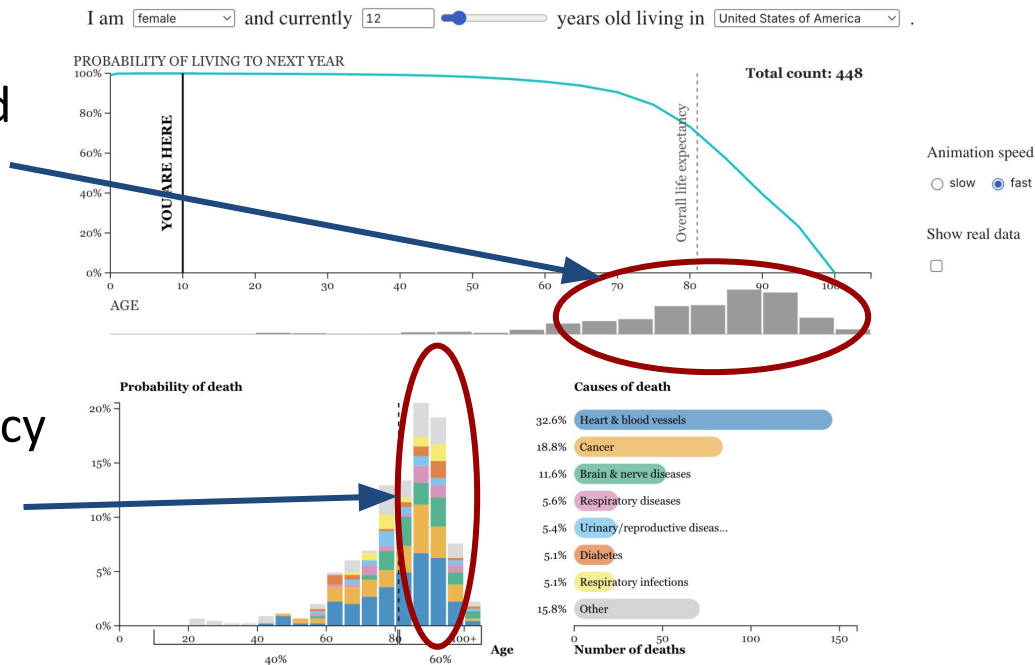
Highlight & Hover (example)



Insights

Life expectancy is only a reference point

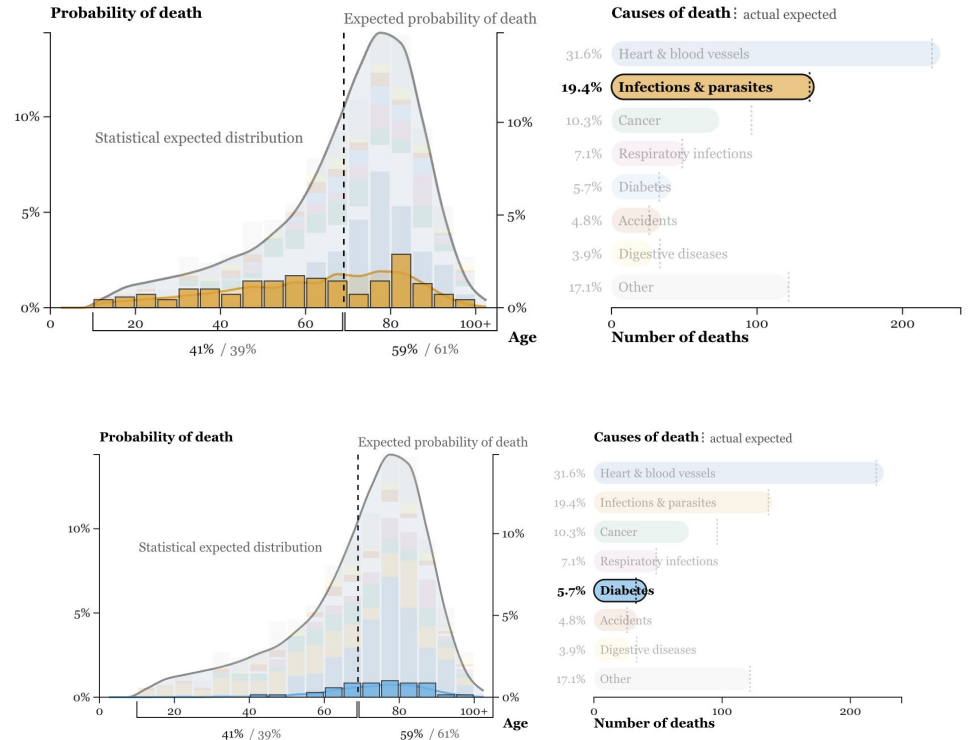
- Possible ages of death spread over a wide range
- Highest probability of death may occur after life expectancy



Insights

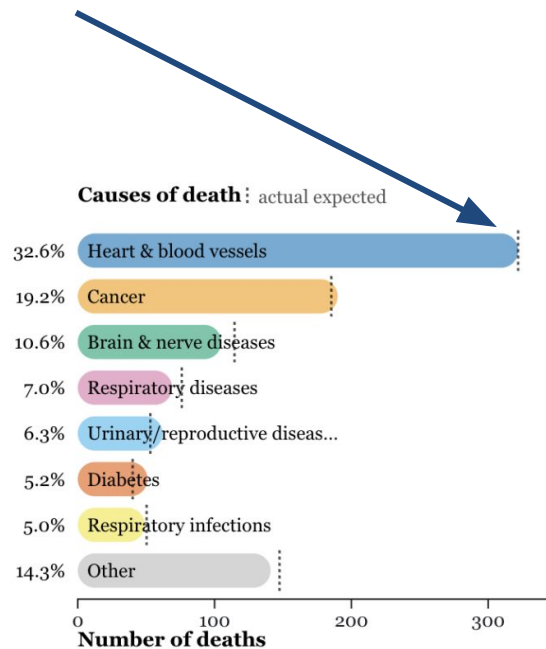
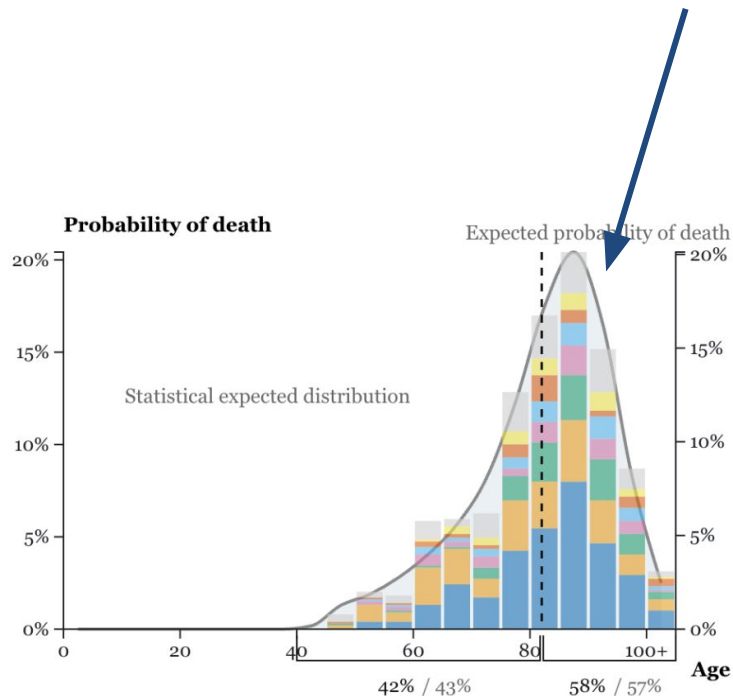
Causes of death differ across life stages

- Some causes appear across almost all ages
- Others are concentrated in specific stages of life.



Insights

Individual randomness accumulates into population patterns



Limitations & future work

- Not practical to display causes of death clearly
 - Future: allow expanding or drilling down into “Other”
- Results are not truly country-specific
 - WHO data only available for 5 aggregate regions. Selected countries mapped to those regions
 - Future: use country-level data if available

Limitations & future work

- Had to use 2019 cause-of-death data
 - Latest WHO data heavily influenced by COVID
 - Future: use post-COVID data when available for more accurate patterns

Contributions

- Connects population data to individual experience
- Makes abstract data intuitive
- Helps understand uncertainty in ages and causes of death
- Helps understand distribution of ages and causes of death
- Shows how randomness forms stable patterns